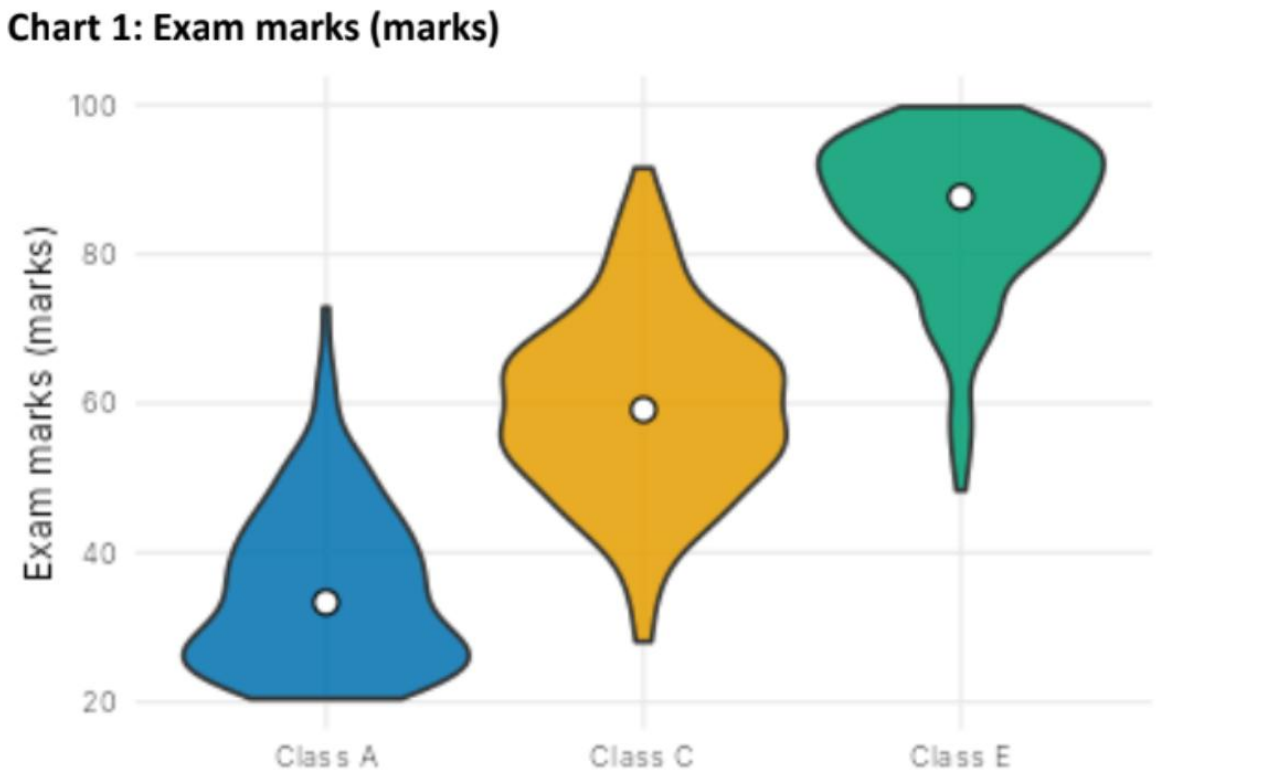
DATA VISUALIZATION USING R

DSA0602 Data Handling and Visualization
CO2 -ACTIVITY-4: Identify Skewness using Violin Plots| Slot D
Name: Jagarajupalli Bhanuprakash Reddy
Reg. No: 192524171
R.No: 15

PART A – Identify Skewness Using Violin Plots

Question (18 marks): Each chart is a violin plot made in R with ggplot2. The white dot represents the median. For every violin, identify the skew and the relation of the mean to the median.

Chart 1: Exam Marks (marks)

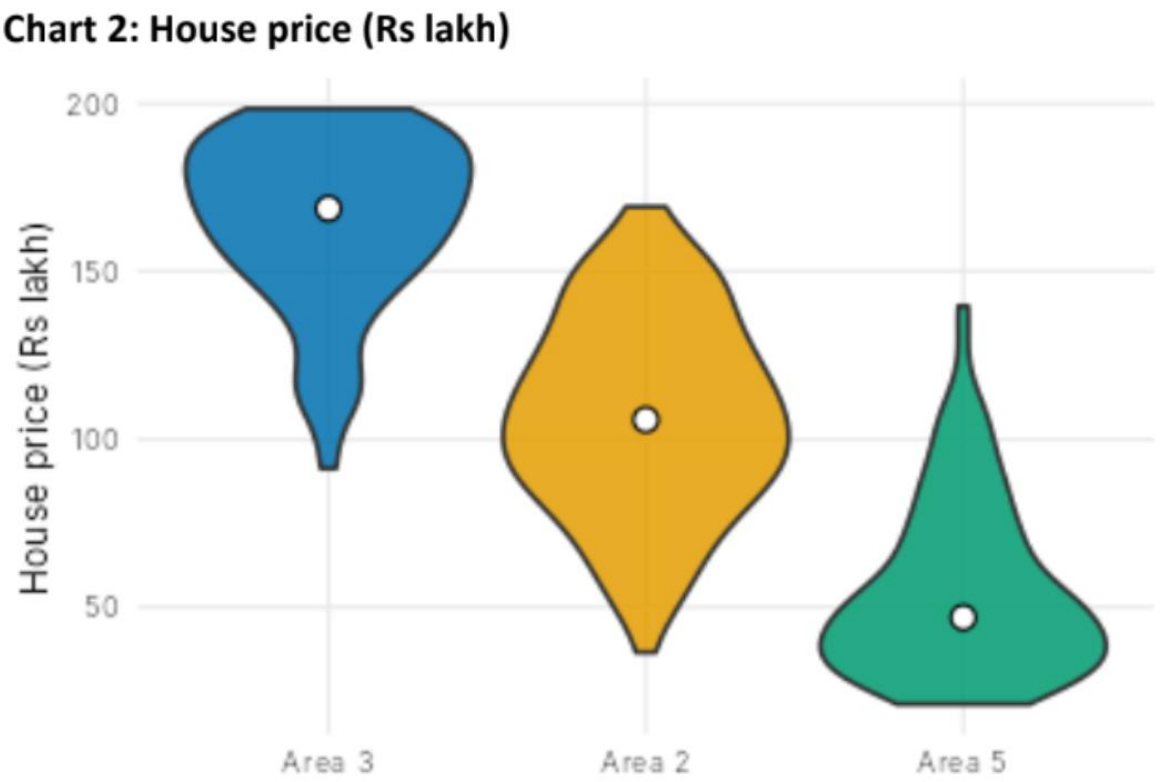


Violin	Skew (circle one)	Mean vs Median (circle)
Class A	Right / Left / Symm.	Mean > / < / ~ Median
Class C	Right / Left / Symm.	Mean > / < / ~ Median
Class E	Right / Left / Symm.	Mean > / < / ~ Median

Violin / Group	Skewness	Mean compared with median
Class A	Right-skewed	Mean > Median
Class C	Right-skewed (slight)	Mean > Median
Class E	Left-skewed	Mean < Median

Interpretation: Class A and Class C have longer tails toward higher marks, indicating positive (right) skew. Class E has a longer tail toward lower marks, indicating negative (left) skew.

Chart 2: House Price (Rs lakh)

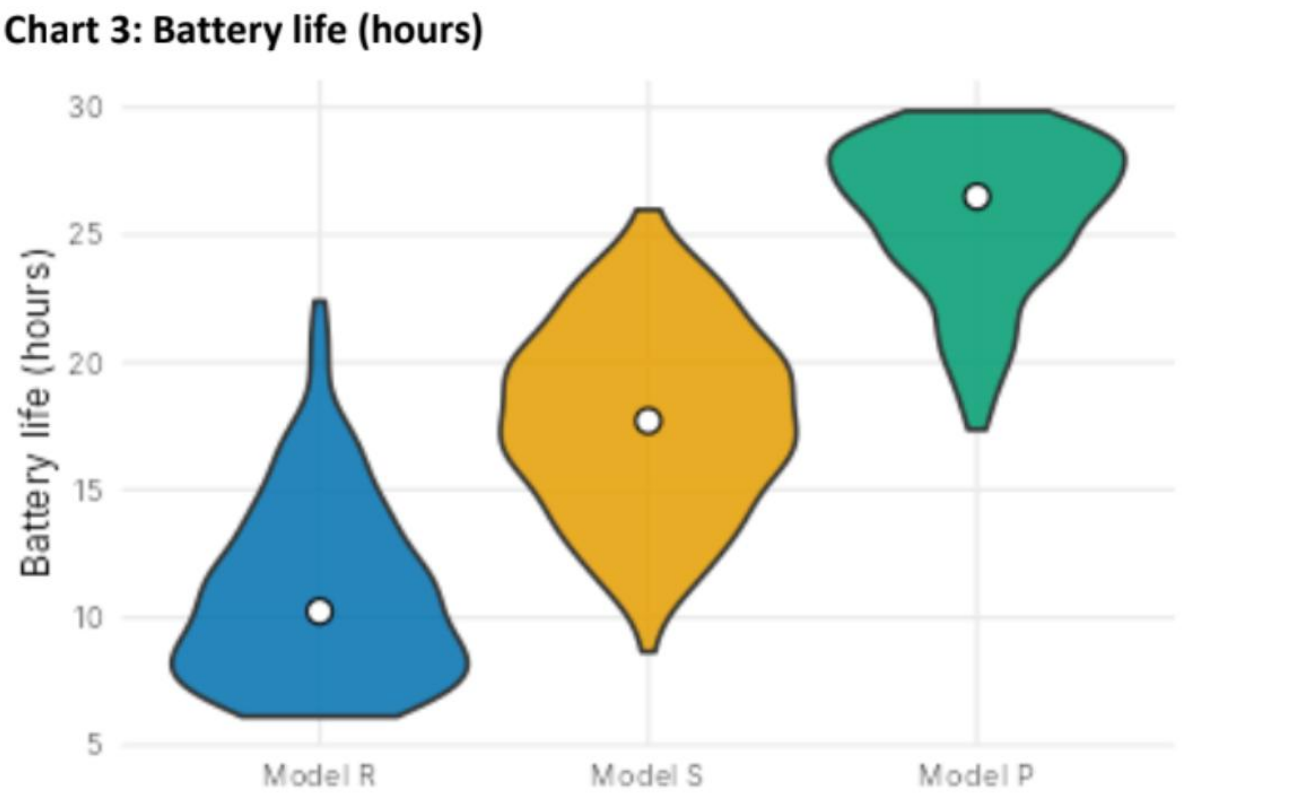


Violin	Skew (circle one)	Mean vs Median (circle)
Area 3	Right / Left / Symm.	Mean > / < / ~ Median
Area 2	Right / Left / Symm.	Mean > / < / ~ Median
Area 5	Right / Left / Symm.	Mean > / < / ~ Median

Violin / Group	Skewness	Mean compared with median
Area 3	Left-skewed	Mean < Median
Area 2	Approximately symmetric	Mean ≈ Median
Area 5	Right-skewed	Mean > Median

Interpretation: Area 3 has a tail toward lower prices; Area 2 is approximately balanced around its centre; Area 5 has a longer tail toward higher prices.

Chart 3: Battery Life (hours)

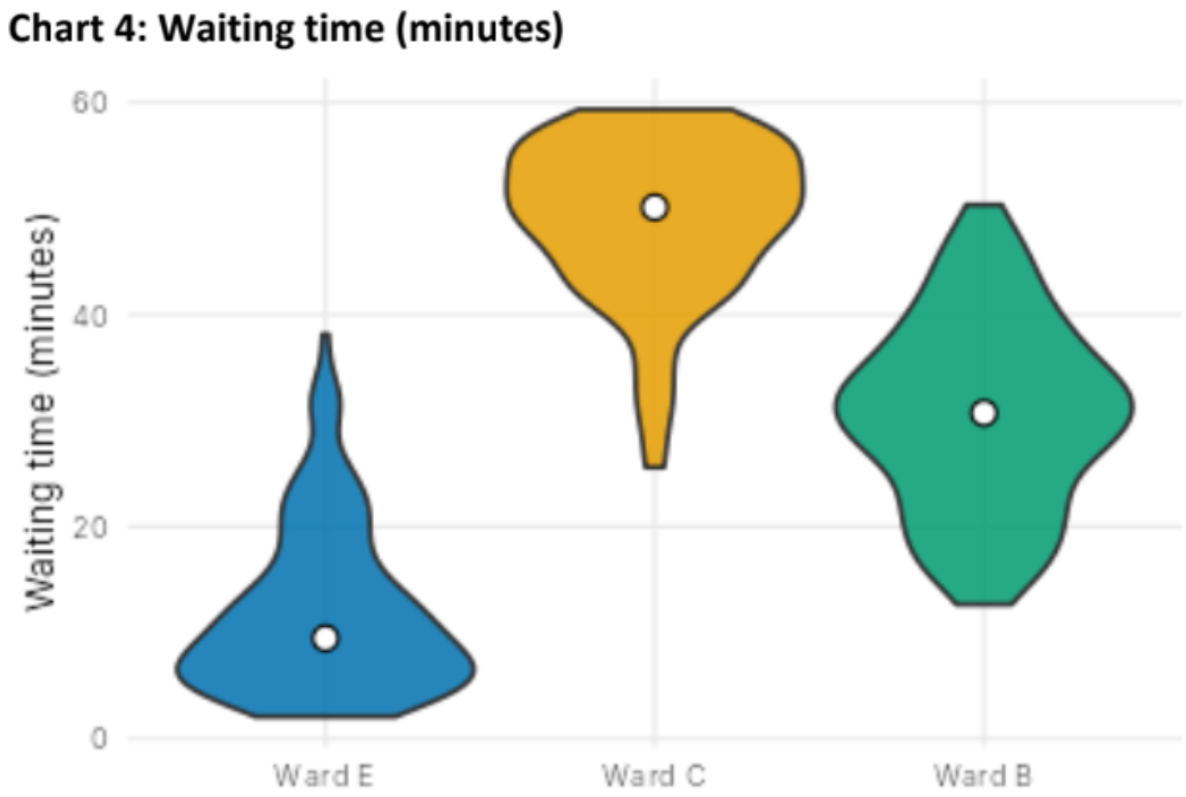


Violin	Skew (circle one)	Mean vs Median (circle)
Model R	Right / Left / Symm.	Mean > / < / ~ Median
Model S	Right / Left / Symm.	Mean > / < / ~ Median
Model P	Right / Left / Symm.	Mean > / < / ~ Median

Violin / Group	Skewness	Mean compared with median
Model R	Right-skewed	Mean > Median
Model S	Approximately symmetric	Mean ≈ Median
Model P	Left-skewed	Mean < Median

Interpretation: Model R has a tail toward longer battery life, Model S is approximately symmetric, and Model P has a tail toward shorter battery life.

Chart 4: Waiting Time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Ward E	Right / Left / Symm.	Mean > / < / ~ Median
Ward C	Right / Left / Symm.	Mean > / < / ~ Median
Ward B	Right / Left / Symm.	Mean > / < / ~ Median

Violin / Group	Skewness	Mean compared with median
Ward E	Right-skewed	Mean > Median
Ward C	Left-skewed	Mean < Median
Ward B	Approximately symmetric	Mean ≈ Median

Interpretation: Ward E has a long tail toward larger waiting times, Ward C has a tail toward shorter waiting times, and Ward B is approximately balanced.

PART B – Short Answers (2 marks)

B1. In Chart 2, violin "Area 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical house price here: mean or median? Give one reason. [1]

Answer: Area 3 has a tail toward the lower house-price values (left-skewed). The median is the better measure of a typical house price because it is less affected by the skewed tail and unusually low values than the mean.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Answer:

```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point", shape = 21, fill = "white")
```

Explanation: geom_violin() displays the distribution for each group, while stat_summary() adds a white point at the median.

Conclusion

A right-skewed distribution has a longer tail toward larger values and usually has mean > median. A left-skewed distribution has a longer tail toward smaller values and usually has mean < median. For an approximately symmetric distribution, mean and median are generally close. The skewness classifications above are visual estimates based on the supplied violin plots.